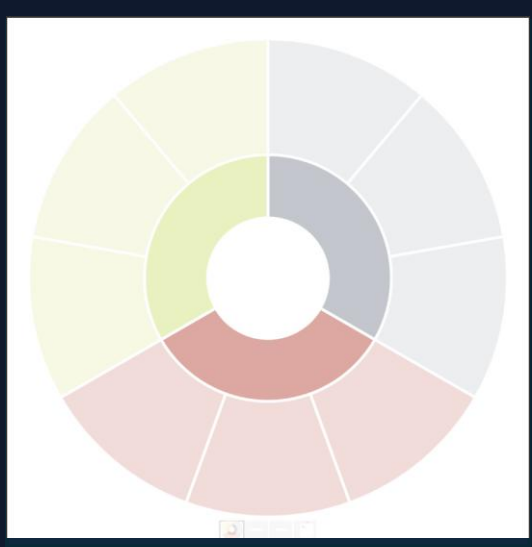




CHAARM-AI

Institution-Specific LLM-Based Intelligent Question Answering System

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PROBLEM & CURRENT SITUATION

Industry Problem: Growing volumes of institutional and technical documents make efficient access to relevant information increasingly difficult.

Current Challenges: Keyword-based search methods often misses contextual meaning.

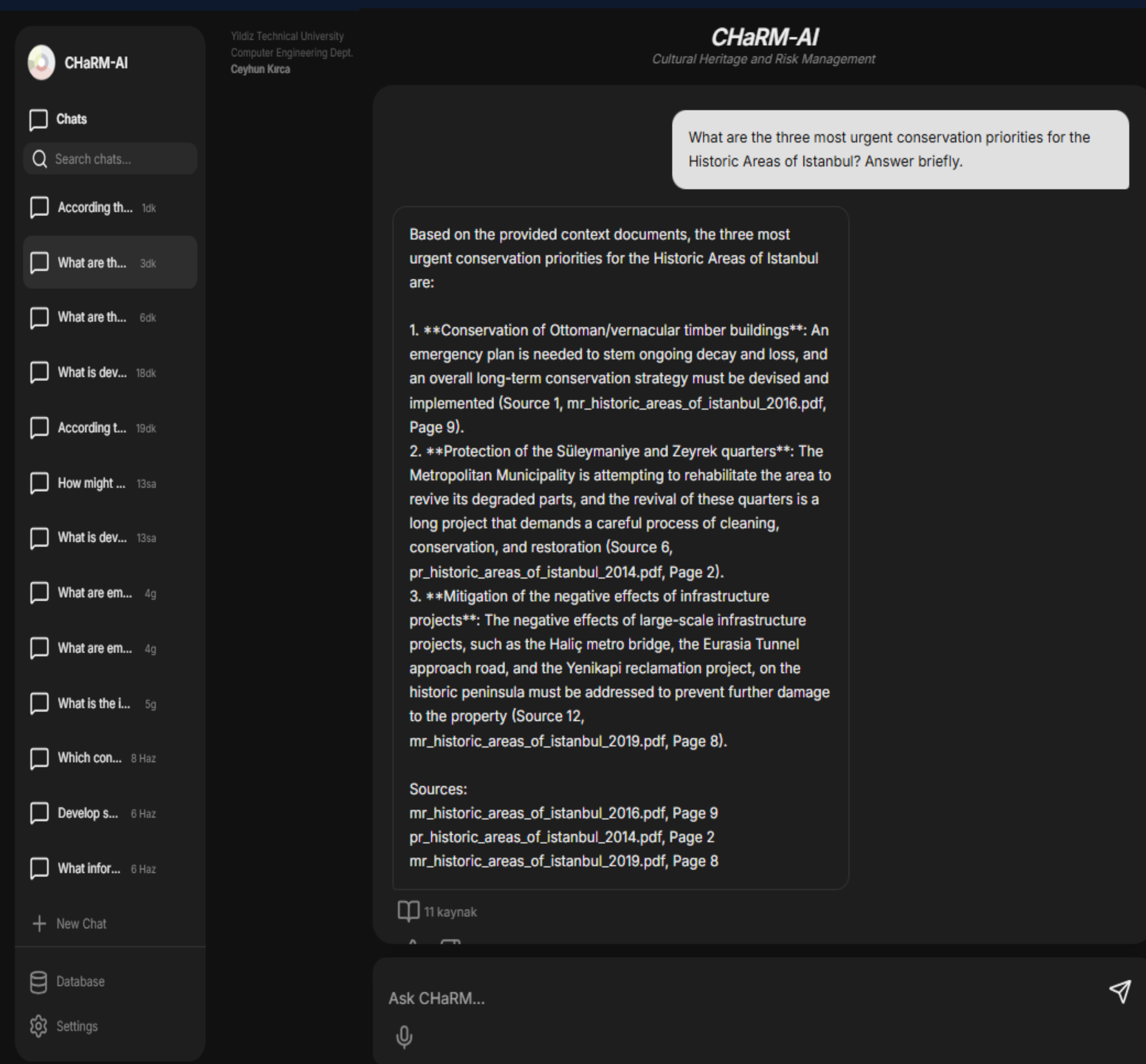
Objective and Need : Develop a reliable question answering system that retrieves relevant content and provides source-supported responses.

PROPOSED SOLUTION & INNOVATION

Core Approach: A RAG-based question answering system combining semantic retrieval and response generation.

Innovation: Generates answers using retrieved document context instead of relying only on model knowledge.

Added Value : Faster information access, reduced manual search effort, and reliable source-supported responses.



Question

What are three most urgent conversation priorities for the Historic Areas of Istanbul?

Answer

1. Conservation of Ottoman/vernacular timber buildings(Source 1,Page 9)
2. Protection of the Süleymaniye and Zeyrek quarters(Source 6, Page 2)
3. Mitigation of the negative effects of infrastructure projects(Source 12, Page 8)

Sources

- 1) mr_historic_areas_of_istanbul_2016,Page 9
- 6) pr_historic_areas_of_istanbul_2014, Page 2
- 12) mr_historic_areas_of_istanbul_2019, Page 8



KEY FINDINGS & CONCLUSION

93.1% Weighted RAGAS Score

- Improved semantic retrieval with two-stage retrieval and reranking
- Reduced hallucinations and increased response reliability
- Fast inference and source-supported answers

CHaRM-AI transformed unstructured cultural heritage documents into an accurate and searchable knowledge base.

